A PROJECT REPORT ON

Kaveri Result Management System

By

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In partial fulfilment of M. Sc. (Computer Science)

Kannada Sangha Pune's

Kaveri College of Arts, Science and Commerce, Pune

AY 2021-22

KANNADA SANGHA PUNE'S

KAVERI COLLEGE OF ARTS, SCIENCE & COMMERCE

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Index No.:ID/PU/PN/C/217/2005 AISHE No.: C-41819 Recognised U/S 2(f) & 12(B) of UGC Act, 1956

NAAC ACCREDITED 'B' GRADE

CERTIFICATE

This is to certify that,

Manisha Narsing Jagtap

has submitted her Industrial project report entitled **"Kaveri Result Management System"** in partial fulfillment of the degree of M.Sc. in Computer Science course to Kannada Sangha's Kaveri College of Arts, Science and Commerce during the academic year 2021-22.

Dr. Ashok M. Agrawal
Principal,
Kaveri College of Arts, Science and Commerce,
Pune.

Acceptance

This manuscript has been read and accepted in satisfaction of the Industrial Training Project requirement of M.Sc (Computer Science) 2021-22, Kannada Sangha’s Kaveri College of Arts, Science and Commerce, Savitribai Phule Pune University.

Examination Committee:

Name	Signature
1.______________________________	______________________
2.______________________________	______________________
3.______________________________	______________________

Place:

Date:

Internship Project

KAVERI RESULT MANAGEMENT SYSTEM

Submitted By:

MANISHA NARSING JAGTAP

ROLL NO: 26

Course: M.Sc. (Computer Science)

Year: Second Year (2021-2022)

Subject: Industrial Training/ Industrial project (CSUTT241)

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1. INTRODUCTION

1.1.Introduction

Kaveri Result Management System is the system where all the aspects related to the proper management of results have done.
These aspects involve managing information about the students, subjects, SGPA, total credits, grade point gained, total and status such as personal details like seat number, name, mother name, Gender, PRN, semester and internal, external, grade and gap respectively organized in a tabular format example Excel file
This system provides an efficient way of managing the results information. Also allows the teachers keep track of the records. This project is based on results marks obtained by student. The first activity is based on reading PDF file provided by the SPPU, read records carefully and keep on adding the record to the system along with the rate which are present in the Grocery

This study is to develop software which manages the results activity done in a college, maintaining the student details, maintaining the records of the student mark obtained for a particular semesters or year. Manual system will make the teachers staff to write calculations and records down which is a long procedure so manual work will increase whereas in proposed system The staff will consume less time in calculation and the entire will be completed within a fraction of seconds so manual entire work will be reduced and teacher can spend more time on the monitoring the result system.

The project will be user friendly and easy to use. The system will display all the records whose status is passed for specified course. Teacher can view the report according to the categories they select

1.Passed ore failed

2. Grade

3. Male or female

Etc.

1.2. Existing System

Manual systems require continuous monitoring to ensure that each record has entered accurately and that students are maintained at the appropriate level. It is also more difficult to share inventory information throughout the system.

Due to this manual process, there could may conflict between Teacher and Student. Each time educator has to go to the PDF Document to check availability of student record and teacher have to place record manually into excel sheet. If customer have existing excel file in the manual system, then educator has to maintain the List hence, manually paper work increases.

A manual result management system relies heavily on the actions of people, which increases the possibility of human error. educator might forget to enter record or simply miscount the number of records. This results in needless additional orders that increase the inventory carrying costs and use up precious storage space. Inaccurate physical counts could also result in not ordering enough of an information, meaning the system could run out of a crucial data at the wrong time.

A manual system has a huge tendency of time wasting as the Teachers could have a lot to tackle while many students seek attention and this is really affecting the management system.

All above points affecting business line and also the important aspect time is consumed.

1.3. Problem Definition

- **Record tracks & concentration:** Keeping track of records is another difficult controlling measure with manual systems. A manual system requires the teacher to write down the record came while passing pdf. This can be a difficult task, as one may lose the list of record or another may forget to write down a sale.

- **Mental and Physical exhaustion:** Over the period of time extreme concentration and focus may cause irritation and frustration for teachers that lead to poor eye sight, back pain and fatigue, dizziness and many more problems

- **Maintain the record list:** Maintaining the student data updating their name and grades and what updates system did that let stakeholder know is tedious job so we are providing computerize detailing once we get update it that will be shown, automatic maintained into system after adding new record.

- **Limitation in record:** A manual system does not update at the end of the day with updated student records.

- **Increase in paper work:** Instead of recording the student detail in manual register staff can handle student itself on system and we are provided soft copy so it reduces the paperwork.

- **Conflicts between conversation among staffs:** In a manual system a pdf divided among themselves so conflict may happen between two or more teacher on same PDF or semester

- **Maintaining records:** Proposed system provides easiest way to maintain records in the database which can be retrieve from the database/excel sheet whenever needed.

- **Time consumption:** This system is mainly focusing on reducing time consumption of pdf parsing and excel entering against huge student records of 60+ pages

2. PROPOSED SYSTEM

2.1. Proposed System

To reduce the shortcomings of the Existing system there is a need to develop a new system that could upgrade the status of the current system which is manual and slow to the system that will be automatic and fast. The new system should be concern with offering the requirements of the faculty, the system should be reliable, easier, fast, and more informative. System also concerns their requirements. It also shows the overall profit.

We have successfully proposed the "Kaveri Result Management System-" for replacing the manual work of the administration/Faculty. This application is flexible and can easily access by the faculty. So, the time taken for getting the information will be reduces

The new system should possess the qualities stated below-

- Reduction in processing cost.
- Error reduction.
- Improve reporting.
- Automatic production of the documents and Reports.
- Faster response time.
- Ability to meet user requirements.
- Flexibility.

- Provides more accuracy.
- Reduced dependency.
- Improves resource uses.
- Reduction in use of the paper.
- Reduction in man power.

- **Main features:**
  - Calculate the Total Marks.
  - Store how many records are there.
  - Store student record and their grades and with other information.
  - Change the Graphical User Interface of the system.

2.2. Purpose

The main aim of the research is to develop a web-based system that will hopefully eliminate or reduce the manual work and also reduce cost and time. We design a computerized Result Management System to maintain certain record level of a of a result management system, such as when to add on more records, keep status and updates of student, thereby helping progress level, record taking and managerial

2.3. Scope

This system work covers stock control, management and tends to correct business. It analyses opening of new stocks, stock updates and ability to view existing ones. It provides quick way of operation by capturing the manual process and automating them. This project is helpful to computerize the item transaction, sales activity record keeping which is a very huge task and maintaining the stock.

Our project has a big scope to do. We can:

- User friendly (as faculties can easily use web-based application).
- ALL time availability. (System remains available as long as computer is well connected with network).
- Easy computation.
- Easy Storage of data.
- More efficient.
- Requires less effort and time.
- Can see the report of the students.
- Change the Graphical User Interface of the system

2.4. Objectives

- To study the functions of college Result Management system.
- To explore the challenges being faced by the manual system.
- The main objective of the Result management system is to reduce the work, for managing the details like course of the student, student record, subject details
- To provide the accuracy for calculating total, grade and credits.
- to reduce ambiguity between multiple records.
- To make a software fast in processing, with good user interface.
- To reduce the hectic processes of search availability of the existing excel sheet.
- Increase the faculty need and loyalty.
- To ensure accurate statistics of student.
- The measure objective of system is to see report to enhance education and focus accordingly.
- To generate history on demand

3. SYSTEM / USER REQUIREMENT ANALYSIS

3.1. Functional Requirements

- The process should consume minimum time to access the whole functionality.
- Faculty must have the authority to control the system and keep track of all records of Students and subject's details.
- The data should be in sorted manner.
- All actors must have a valid login id to enter into the site.
- Account login should be feasible.
- The system should store the information about new entries of student, subject and semester and maintain their quantity record.
- The system should provide the search area, security of data.

3.2. Non-Functional Requirements

- Nullify the process of visiting the any files manually.
- The system shall accommodate high number of records and students without any fault.
- System must be easy to understand and simple to adopt.
- System use shall not cause any harm to human users.
- System shall use secured database.
- Normal users can just read information but they cannot edit or modify anything or any other information.

4. OPERATIONAL ENVIRONMENT

4.1. Feasibility Study

The result of the feasibility study will indicate whether to proceed with the proposed system or not. If the results of the feasibility study are positive, then we can proceed to develop a system otherwise project should not be pursued

4.1.1. Technical Feasibility

The current system developed is technically feasible. It is a web-based user interface. it provides an easy access to the users. The database's purpose is to create, establish and maintain a workflow among various entities. Permission to the users would be granted based on the roles specified. Therefore, it provides the technical guarantee of accuracy, reliability and security. The software and hardware requirements for the development of this project are not many and are already available as free as open source.

A study of resources availability that may affect the ability to achieve an acceptable system. This evaluation determines whether the technology needed for the proposed system is available or not. This system can be made in any language that support good user interface and easy database handling.

- **Hardware Requirements:**
  - Basic configured device

- **software Requirements:**
  - Operating system:

    Windows: 7, 8, 8.1, 10,11 but have to install

    Linux: all version of Linux operating system

    Mac: we can also configure this project on Mac if needed
  - Front End: HTML, CSS.
  - Backend: MySQL.
  - Browser: chrome.
  - Web server: Apache tomcat.
  - Software Development kit: JDK.
  - Eclipse is an integrated development environment (IDE) used in computer programming
  - Scripting language Enable: JavaScript.
  - Database JDBC Driver: MySQL JConnector

4.1.2. Operational Feasibility

Operational feasibility of the project is to be taken as an important part of the project implementation. Operational feasibility deals with operational efficiency of a system.

The project is beneficial only if they can be turned out into information system. The management issues and user requirements have been taken into consideration while development. The system can be used and work properly if it is being developed and implemented and would help in the improvement of performance status

5. MODULE LIST

We will use 5 modules in this project. These are as follows:

- **Student:** Student module is major module into this application, which has attributes like Faculties are going to put his/her information into tabular format and save it as record as well as generate further reports

- **Subject:** Inside subject module we will keep track of internal marks, external marks, subject code, Gap, tutorial marks and Total marks.

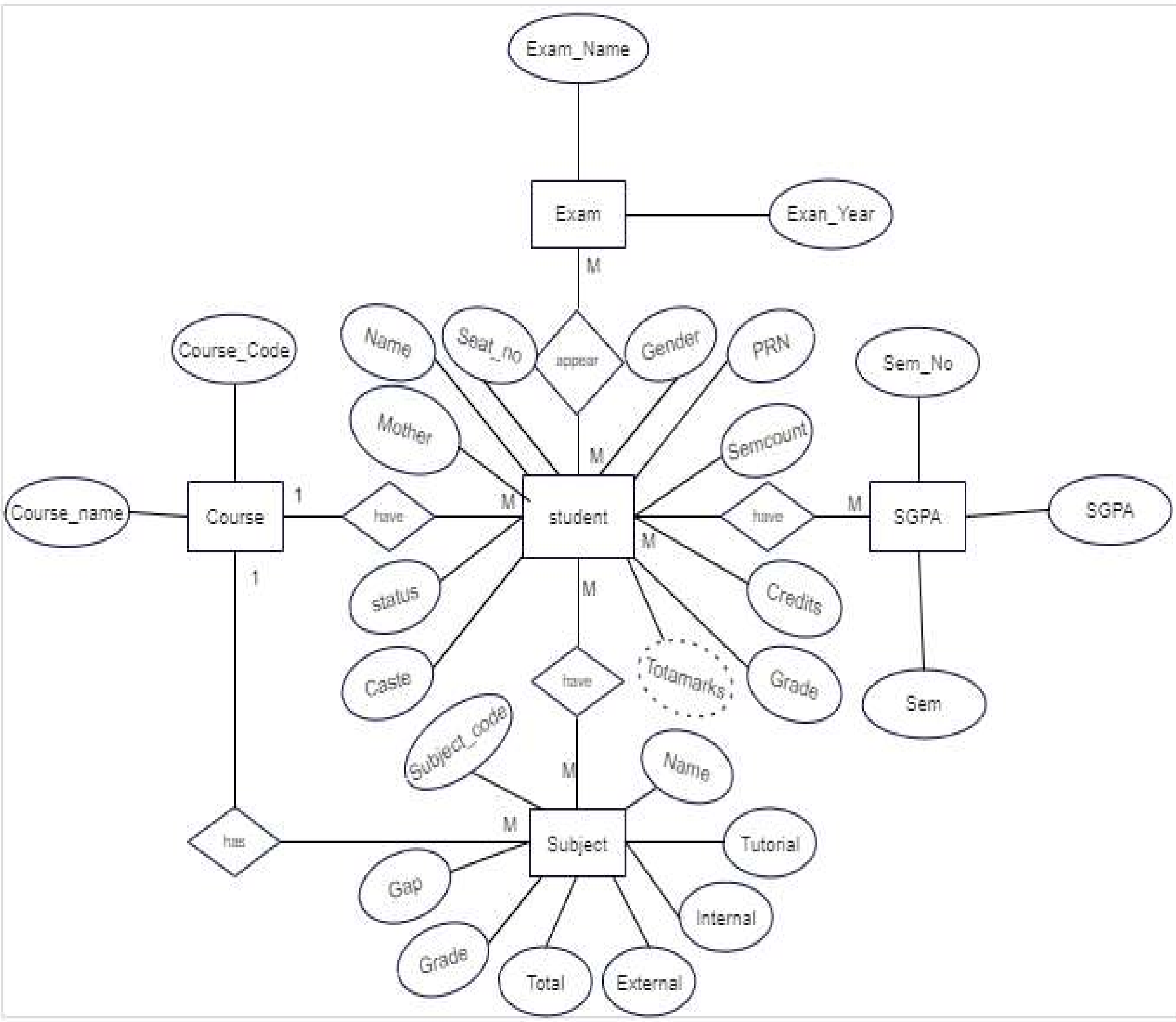
- **course:** inside course module will trace which branch is these, hence we can segregate student, Subject and rest module accordingly

- **SGPA:** In SGPA module, we will store semester number and SGPA got into that corresponding semester

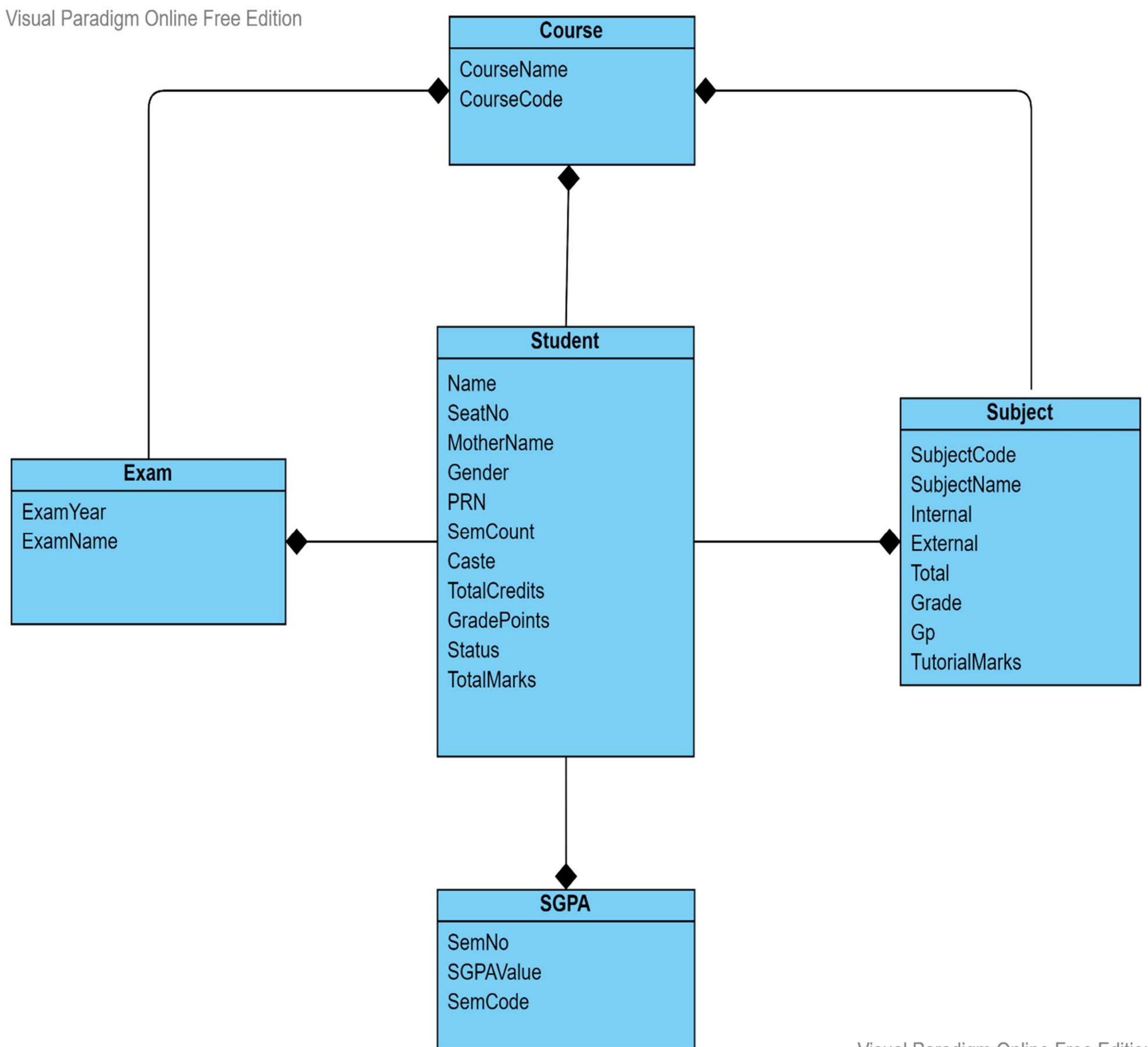
- **Exam:** In Exam module we will keep the track of Exam Names and Exam Year if any teachers want to take look at excel sheet of that particular exam years/ exam name they can easily access it.

6. DIAGRAM

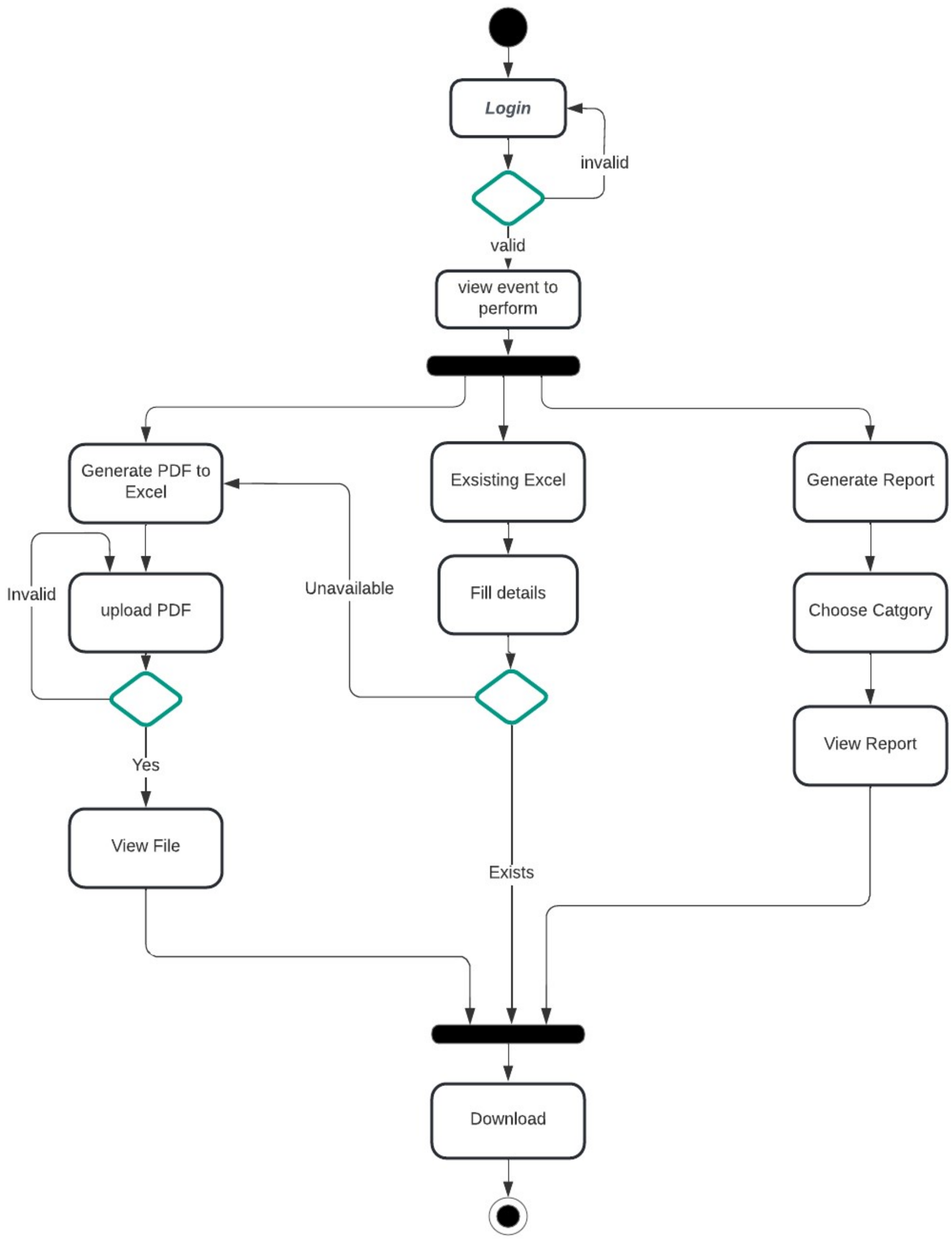
6.1. ER-Diagram



6.2. Class Daigram:

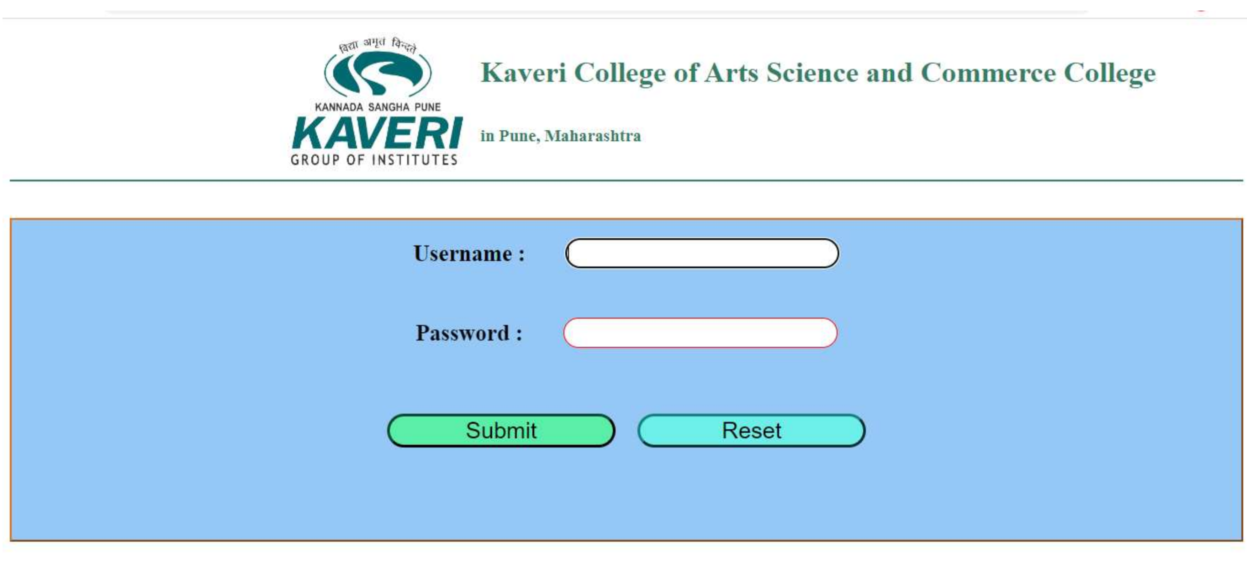


6.3. Activity Diagram:

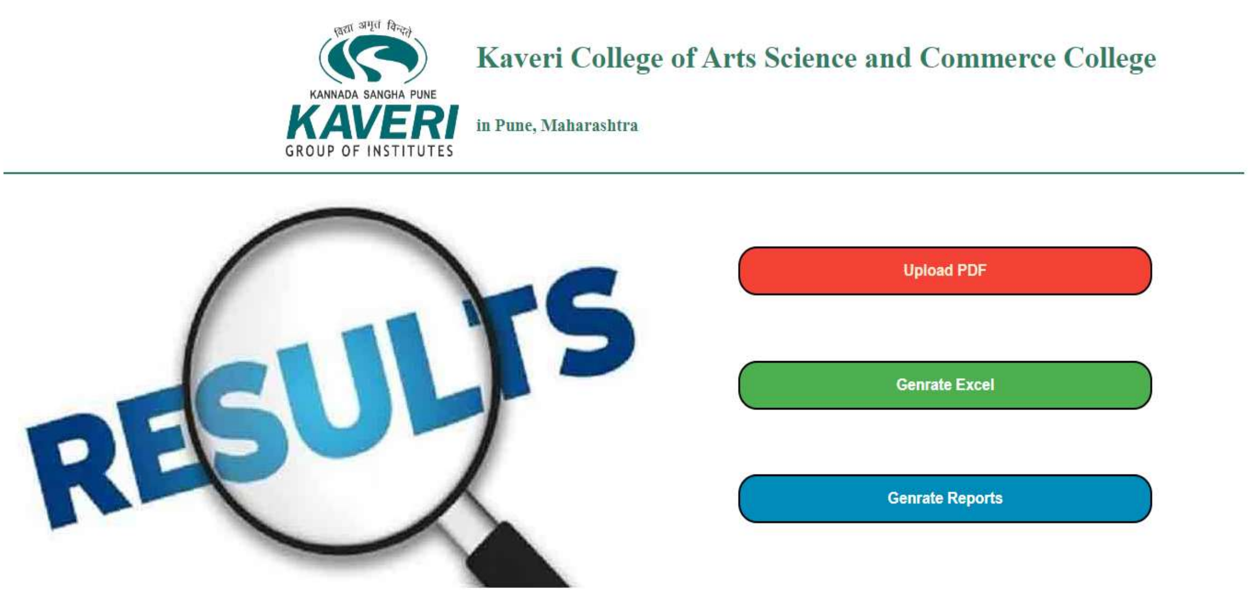


7. INPUT & OUTPUT SCREENS

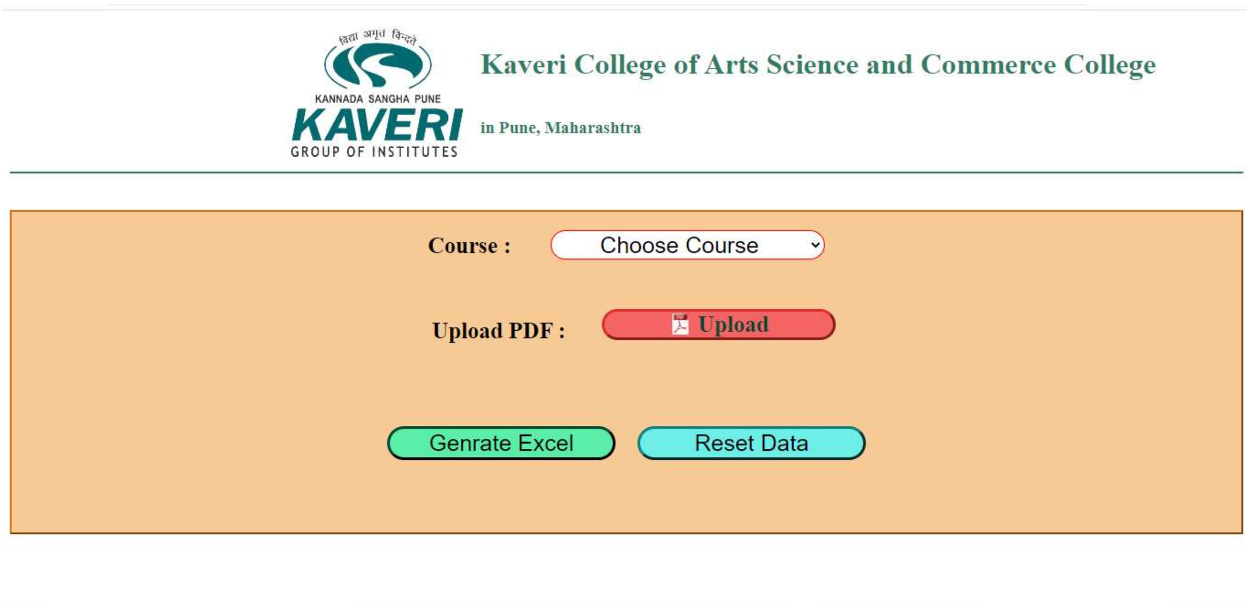
1. Login



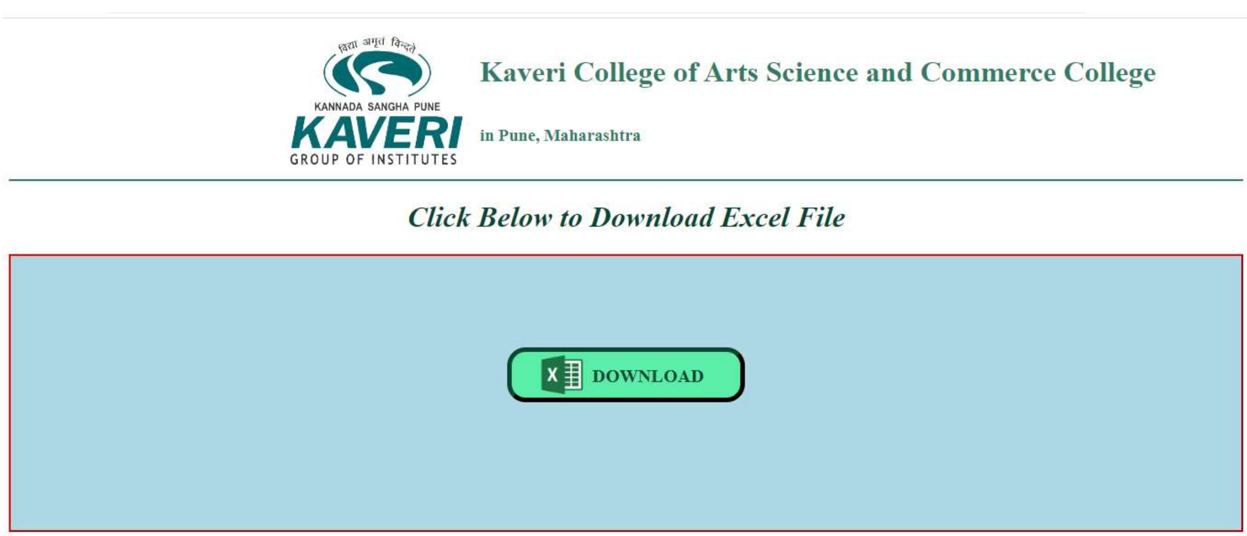
2. Homepage



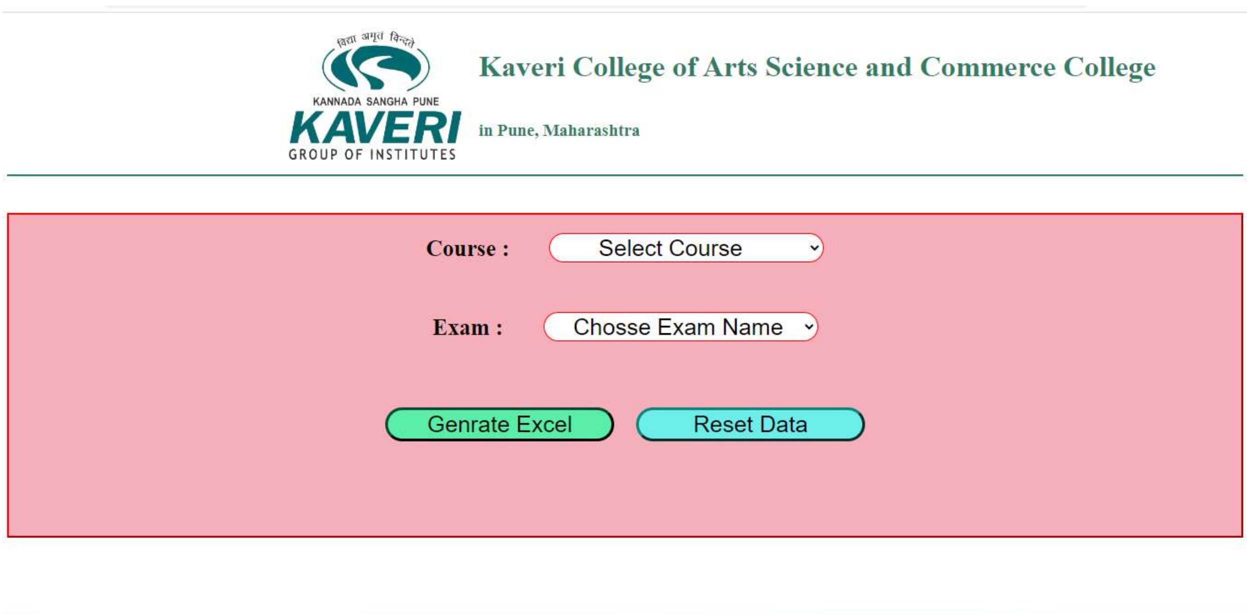
3. Upload PDF

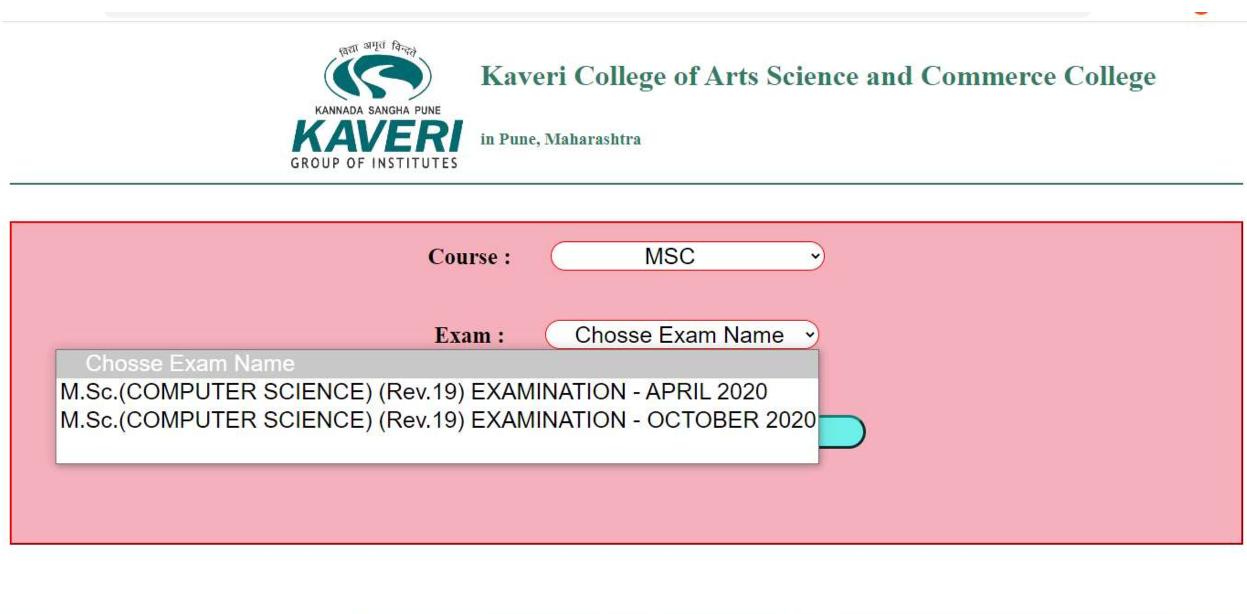


4. Generate Excel

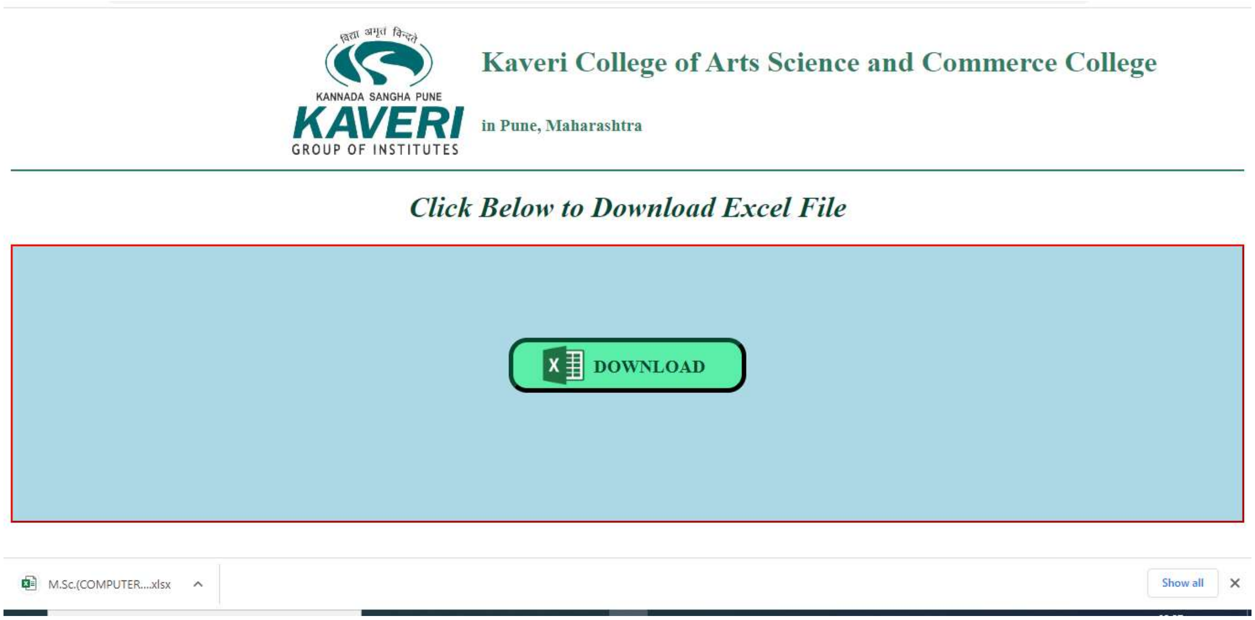


5. Existing Excel

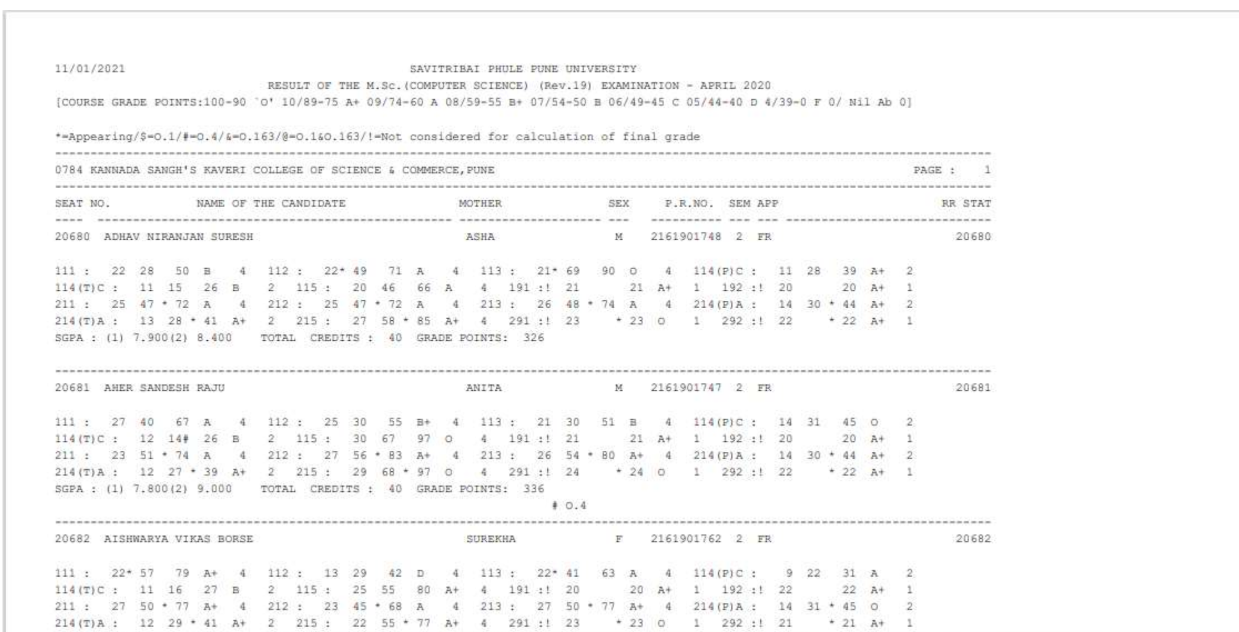




6. Download Excel file



Input-File



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11/01/2021                                   SAVITRIBAI PHULE PUNE UNIVERSITY
                              RESULT OF THE M.Sc.(COMPUTER SCIENCE) (Rev.19) EXAMINATION - APRIL 2020
[COURSE GRADE POINTS:100-90 `O' 10/89-75 A+ 09/74-60 A 08/59-55 B+ 07/54-50 B 06/49-45 C 05/44-40 D 4/39-0 F 0/ Nil Ab 0]

*=Appearing/$=0.1/#=0.4/&=0.163/@=0.1&0.163/!=Not considered for calculation of final grade
-----------------------------------------------------------------------------------------------------------------------------
0784 KANNADA SANGH'S KAVERI COLLEGE OF SCIENCE & COMMERCE,PUNE                                                   PAGE :     1
-----------------------------------------------------------------------------------------------------------------------------
SEAT NO.          NAME OF THE CANDIDATE                   MOTHER               SEX     P.R.NO.  SEM APP                RR STAT
---- -------------------------------------------------- -------------------- ---   ---------- --- --- -------------------------
20680  ADHAV NIRANJAN SURESH                               ASHA                 M    2161901748  2  FR                    20680

111 :   22  28   50  B    4   112 :   22* 49   71  A    4   113 :   21* 69   90  O    4   114(P)C :   11  28   39  A+   2
114(T)C :   11  15   26  B    2   115 :   20  46   66  A    4   191 :! 21       21  A+   1   192 :! 20       20  A+   1
211 :   25  47 * 72  A    4   212 :   25  47 * 72  A    4   213 :   26  48 * 74  A    4   214(P)A :   14  30 * 44  A+   2
214(T)A :   13  28 * 41  A+   2   215 :   27  58 * 85  A+   4   291 :! 23     * 23  O    1   292 :! 22     * 22  A+   1
SGPA : (1) 7.900(2) 8.400    TOTAL  CREDITS :  40  GRADE POINTS:  326

-----------------------------------------------------------------------------------------------------------------------------
20681  AHER SANDESH RAJU                                   ANITA                M    2161901747  2  FR                    20681

111 :   27  40   67  A    4   112 :   25  30   55  B+   4   113 :   21  30   51  B    4   114(P)C :   14  31   45  O    2
114(T)C :   12  14#  26  B    2   115 :   30  67   97  O    4   191 :! 21       21  A+   1   192 :! 20       20  A+   1
211 :   23  51 * 74  A    4   212 :   27  56 * 83  A+   4   213 :   26  54 * 80  A+   4   214(P)A :   14  30 * 44  A+   2
214(T)A :   12  27 * 39  A+   2   215 :   29  68 * 97  O    4   291 :! 24     * 24  O    1   292 :! 22     * 22  A+   1
SGPA : (1) 7.800(2) 9.000    TOTAL  CREDITS :  40  GRADE POINTS:  336
                                                                  # 0.4
-----------------------------------------------------------------------------------------------------------------------------
20682  AISHWARYA VIKAS BORSE                               SUREKHA              F    2161901762  2  FR                    20682

111 :   22* 57   79  A+   4   112 :   13  29   42  D    4   113 :   22* 41   63  A    4   114(P)C :    9  22   31  A    2
114(T)C :   11  16   27  B    2   115 :   25  55   80  A+   4   191 :! 20       20  A+   1   192 :! 22       22  A+   1
211 :   27  50 * 77  A+   4   212 :   23  45 * 68  A    4   213 :   27  50 * 77  A+   4   214(P)A :   14  31 * 45  O    2
214(T)A :   12  29 * 41  A+   2   215 :   22  55 * 77  A+   4   291 :! 23     * 23  O    1   292 :! 21     * 21  A+   1
```

Output File:

Student Details						111					112					113
Seat No.	Name	Mother Name	Gender	PRN	Semester	INT	EXT	TOT	Grade	Gp	INT	EXT	TOT	Grade	Gp	INT
20680	ADHAV NIRANJAN SURESH	ASHA	M	2161901748	2	22	28	50	B	4	22	49	71	A	4	
20681	AHER SANDESH RAJU	ANITA	M	2161901747	2	27	40	67	A	4	25	30	55	B+	4	
20682	AISHWARYA VIKAS BORSE	SUREKHA	F	2161901762	2	22	57	79	A+	4	13	29	42	D	4	
20683	AKASH GANESH TANPURE	SHAILA	M	2161901761	2	22	28	50	B	4	25	28	53	B	4	
20684	AMRUTKAR SHIVANI DILIP	PRATIBHA	F	2161901743	2	26	31	57	B+	4	25	32	57	B+	4	
20685	BHUS MAYUR SANJAY	JYOTI	M	2161901741	2	23	41	64	A	4	21	29	50	B	4	
20686	BIRWATKAR RUSHIKESH VIDYADI	VIJAYA	M	2161901766	2	22	28	50	B	4	12	52	64	A	4	
20687	GADADE PRAMOD RAJKUMAR	SAVITA	M	2161901764	2	25	28	53	B	4	25	30	55	B+	4	
20688	KARANJKAR DAMINI PRITHVIRAJ	DEVI	F	2161901759	2	25	28	53	B	4	22	28	50	B	4	
20689	KHAWALE ASHWINI RAMESHRAC	SANGEETA	F	2161901765	2	26	45	71	A	4	25	35	60	A	4	
20690	PAI RAHUL VENKATDAS	SARASWATI	M	2161901738	2	28	28	56	B+	4	26	28	54	B	4	
20691	PASALKAR SONAL RAVINDRA	RESHMA	F	2161901752	2	25	32	57	B+	4	28	32	60	A	4	
20692	PAWAR KIRTI ASHOKRAO	SANDHYA	F	2161901760	2	24	49	73	A	4	26	28	54	B	4	
20693	PAYGUDE VRUSHALI ANIL	RADHIKA	F	2161901750	2	25	35	60	A	4	28	33	61	A	4	
20694	SANKPAL RUPESH ANIL	KALPANA	M	2161901740	2	22	29	51	B	4	26	28	54	B	4	
20695	SHINDE ANKUR PRAKASH	KAVITA	M	2161901767	2	22	40	62	A	4	14	28	42	D	4	
20696	SUTAR VAISHNAVI DEEPAK	SAVITA	F	2161901758	2	25	38	63	A	4	22	30	52	B	4	
20697	THADKE AJAY VYANKATRAO	KALPANA	M	2161901749	2	22	29	51	B	4	25	28	53	B	4	

			291			292			SGPA		Final Result			
	Grade	Gp	INT	Grade	Gp	INT	Grade	Gp	SGPA(1)	SGPA(2)	Total Credits	Grade Points	Total	Status
27	A+	4	23	O	1	22	A+	1	7.9	8.4	40	326	342	PASS
29	O	4	24	O	1	22	A+	1	7.8	9	40	336	341	PASS
22	A+	4	23	O	1	21	A+	1	7.4	8.9	40	326	322	PASS
26	O	4	23	O	1	19	A+	1	7.4	9	40	328	318	PASS
29	O	4	24	O	1	22	A+	1	8.4	9	40	348	365	PASS
15	A	4	20	A+	1	15	A	1	7.9	8	40	318	340	PASS
15	B+	4	23	O	1	19	A+	1	7.6	7.4	40	300	301	PASS
24	A+	4	23	O	1	22	A+	1	7.2	8.8	40	320	296	PASS
21	A	4	23	O	1	20	A+	1	7.1	8	40	302	312	PASS
28	A+	4	21	A+	1	22	A+	1	7.9	8.8	40	334	315	PASS
27	O	4	22	A+	1	22	A+	1	8.3	9.2	40	350	371	PASS
28	O	4	22	A+	1	22	A+	1	8	9.2	40	344	338	PASS
25	A+	4	23	O	1	19	A+	1	7.3	8.4	40	314	317	PASS
29	O	4	23	O	1	22	A+	1	8.7	8.8	40	350	372	PASS
17	A	4	21	A+	1	22	A+	1	7.6	8.2	40	316	335	PASS
20	A+	4	24	O	1	19	A+	1	7.2	8.2	40	308	319	PASS
20	A	4	23	O	1	20	A+	1	7.8	8.4	40	324	324	PASS
25	A+	4	23	O	1	22	A+	1	6.9	8.4	40	306	294	PASS

8. IMPLEMENTATION VIEW

FileInputStream: class is use to open and read a file from a file in the form of sequence of bytes. File as input.

org.apache.pdfbox: The Apache PDFBox library is an open source Java tool for working with PDF documents.

PDDocument: Creates an empty PDF document.

need to add at least one page for the document to be valid.

Load an existing PDF document using the static method load() of the PDDocument class

PDFTextStripper: This class will take a pdf document and strip out all of the text and ignore the formatting and such

org.apache.poi: Apache POI is a popular API that allows programmers to create, modify, and display MS Office files using Java programs.
to use HSSF and XSSF read and write spreadsheets purpose

XSSFWorkbook: It is a class that is used to represent both high-level and low-level Excel file formats. It belongs to the org so one can create Excel sheet, rows and columns

Array List: to store student data subject data we are using array list class uses a dynamic array for storing the elements.
It is like an array, but there is no size limit. We can add or remove elements anytime.

Hash Map: which allows us to store key and value pair, where keys should be unique. So, for each unique Column name there will be column number assigned to it.

9. Conclusion Remark

9.1. Limitations of the System:

- Normal user like faculty cannot edit or modify after score entered into database but they could do change after downloading excel.
- This System is totally made for faculty which limit system where student cannot access.
- If there is problem or error exists in PDF file which SPPU has provided so it may lead to lose of student data.
- Points may vary as per project goes onward.

9.2. Future Scope:

- We can let student access the result
- Faculty can send the particular report directly to higher authority person within system.
- Points may vary as per project goes onward

9.3. Conclusion:

It is concluded that the system will works well and thus it will fulfil the end user's requirement. The system is tested and errors are accurately removed. This application will be accessed from one system and hence login from that one system is tested. This system is user friendly so that everyone can use this application easily. Faculty or staff has to give the proper documentation.

The staff is able to easily understand how this overall application. The system is evaluated, implemented and its performance is found to be satisfactory to the end users.

The required result for the user's requirements is generated. Further enhancements can be added to this system, because the features of this application are very attractive and it is useful than the present one

10. References

https://online.visual-paradigm.com/

https://stackoverflow.com/

https://mvnrepository.com/

https://www.edrawmax.com/

You-tube channels